

Polynomial Regression with Regularization

Roll No. BT2024081 | Datasets: var1 (6 features) and var2 (3 features)

Code: https://github.com/Srikar-V06/Polynomial_Regression

1. Problem and Approach

Each dataset has 1000 samples and a continuous target y . var1 has six inputs (x_1 – x_6 , shape 1000×7 with target) and var2 has three (x_1 – x_3 , shape 1000×4). Inputs lie in $[-1, 1]$, and many values sit exactly at ± 1 , which suggests the features were clipped to that range. A linear model is clearly insufficient (degree-1 $R^2 = 0.10$ on var1 and 0.27 on var2), so the task is to capture non-linear structure with polynomial feature expansion while controlling the variance this introduces.

The same pipeline was used for both problems:

- **Feature expansion:** `PolynomialFeatures(degree=d, include_bias=False)`, which adds all powers and interaction terms between variables (the intercept is handled by the regressor).
- **Standardization:** a `StandardScaler` is applied *after* the polynomial expansion, so every polynomial term (x , x^2 , $x_1 \cdot x_2$, ...) has zero mean and unit variance. Without this, the L1/L2 penalty would punish high-order terms unevenly because of their very different scales.
- **Regularized linear models:** Ridge (L2) and Lasso (L1), since high-degree expansions produce hundreds to thousands of correlated terms and plain least squares would overfit.
- **Grid search:** $\text{Model} \in \{\text{Ridge}, \text{Lasso}\}$, $\alpha \in \{0.01, 0.1, 1\}$, degree 1–10 for var1 (60 configurations) and 1–13 for var2 (78 configurations).
- **Selection by cross-validation:** every configuration was scored by 5-fold CV (shuffled, `random_state=42`) on MSE and R^2 . The configuration with the lowest mean CV MSE was selected, refit on all 1000 training rows, and used to predict the test set. Lasso used `max_iter=10000`. Because expansion and scaling sit inside the pipeline, they are re-fit within each fold, so there is no leakage from validation data.
- **Pipeline design:** the whole `expand` $\rightarrow$ `scale` $\rightarrow$ `regress` chain is one scikit-learn Pipeline, so the same code ran both datasets with only the maximum degree changed (10 for var1, 13 for var2).

2. Final Choices

Dataset	Model	Alpha	Degree	CV MSE	CV R^2
var1	Lasso	0.01	5	0.3394	0.9638
var2	Ridge	1.0	10	0.2409	0.9939

Feature counts explain why the two datasets prefer different degrees: with 6 inputs, degree 5 already gives 461 features and degree 10 gives 8007; with 3 inputs, degree 10 gives only 285 and degree 13 gives 559. Higher degrees are far cheaper in variance on var2.

3. Dataset var1

Chosen: Lasso, $\alpha = 0.01$, degree 5 (CV MSE 0.339, R^2 0.964).

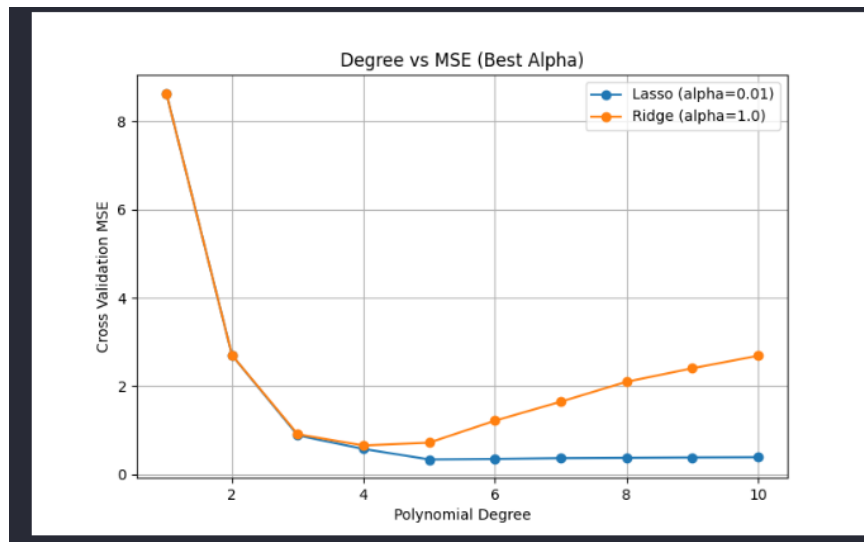


Figure 1. Cross-validated MSE vs. polynomial degree for the best α of each model (var1).

Degree	Ridge ($\alpha=1$) MSE	Lasso ($\alpha=0.01$) MSE
1	8.635	8.636
2	2.700	2.698
3	0.910	0.885
4	0.656	0.577
5	0.720	0.339
6	1.217	0.351
8	2.098	0.376
10	2.687	0.390

Rationale

- **Underfitting below degree 4.** MSE drops from 8.6 ($d=1$) to 2.7 ($d=2$) to under 1 at $d=3$, so the target has strong non-linear structure and interactions.
- **Ridge overfits past its optimum.** Ridge ($\alpha=1$) bottoms out at degree 4 (MSE 0.656) and then rises steadily to 2.69 at degree 10. Ridge shrinks coefficients but keeps every term, so the growing feature set adds variance.
- **Lasso stays stable.** Lasso ($\alpha=0.01$) reaches its minimum at degree 5 and then stays nearly flat (0.34–0.39 for degrees 5–10). Its L1 penalty zeroes out irrelevant high-order terms, which suggests the true function is sparse in the polynomial basis.
- **Why degree 5 and not 6–10.** Degrees 5–10 are within about 0.05 MSE of each other, so higher degrees add no measurable accuracy. Degree 5 has the lowest error and is the simplest model, so it is the safest choice for generalization.
- **Regularization strength.** At $\alpha = 1$, Lasso over-shrinks and collapses ($R^2 \approx 0.25$ for degrees 3–10); at $\alpha = 0.1$ it reaches only $R^2 \approx 0.915$. The weakest penalty (0.01) was best, so the grid minimum was selected.
- **Instability at low α .** Ridge with $\alpha = 0.01$ at degree 6 gave $R^2 = -1.56$ (MSE 23.5), a sign of an ill-conditioned design matrix when the penalty is too weak to stabilize correlated high-degree terms.

4. Dataset var2

Chosen: Ridge, $\alpha = 1.0$, degree 10 (CV MSE 0.241, R^2 0.994).

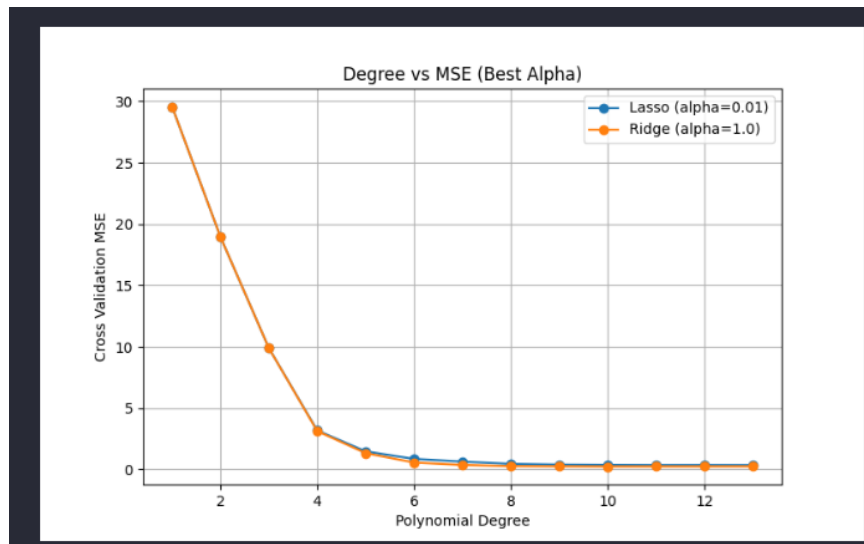


Figure 2. Cross-validated MSE vs. polynomial degree for the best α of each model (var2).

Rationale

- **A much higher-degree relationship.** MSE falls from 29.6 ($d=1$) to 18.95 ($d=2$), 9.89 ($d=3$), 3.12 ($d=4$) and 1.32 ($d=5$), then flattens near 0.2–0.4 by degrees 8–13. The function needs a high-degree polynomial to be captured.
- **No overfitting at high degree.** Unlike var1, neither model's curve turns upward; with only 3 inputs the feature count stays manageable (285 features at $d=10$), and regularization keeps it stable.
- **Ridge vs. Lasso.** The best Ridge ($\alpha=1$, $d=10$) beat the best Lasso ($\alpha=0.01$, $d=13$): 0.241 vs. 0.349 MSE, with R^2 0.9939 vs. 0.9911. Because the target depends on many smooth polynomial terms rather than a few, Ridge's keep-all-and-shrink behavior fits better here than Lasso's sparsity.
- **Why degree 10 and not 13.** The curve has plateaued by degree 10, so the extra degrees would add 274 more features for no gain. Lower-degree models ($d \leq 5$) clearly underfit.
- **Weak Lasso at $\alpha = 1$.** Lasso with $\alpha = 1$ plateaus at $R^2 \approx 0.65$ (MSE ≈ 14), again showing that a strong L1 penalty removes useful terms.

5. Other Techniques and Observations

- **Bias-variance trade-off:** the degree controls model flexibility and the penalty α controls variance; both were tuned jointly rather than separately.
- **Model comparison across L1 and L2:** comparing both penalties revealed that var1 favors sparsity (Lasso) and var2 favors dense shrinkage (Ridge), which a single-model search would have missed.
- **Standardizing polynomial terms:** scaling after expansion keeps the penalty comparable across terms and improves the conditioning of the coordinate-descent solver used by Lasso.
- **Selecting the simplest adequate model:** when several configurations were practically tied (var1 degrees 5–10, var2 degrees 10–13), the lower degree was preferred.
- **Output:** final models were refit and used to predict the 1000 test rows for each dataset (BT2024081_pred_var1.csv, BT2024081_pred_var2.csv).

6. Limitations

The reported R^2 and MSE are 5-fold CV scores on the same data used to pick the model, so they are slightly optimistic; a separate hold-out set or nested CV would give an unbiased estimate. The CV split uses a single fixed seed, so small MSE differences (e.g., var1 degrees 5–7) are within noise. The α grid has only three values, and the best var1 Lasso α sits at the lower edge, so a finer search (e.g., 0.001–0.03) or an ElasticNet could improve it further.